



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-033 BrightBasket Transformation Mini-Lab

Integrated limited-hint assessment | incremental + history + tested marts

Artifact	C3_033_BRIGHTBASKET_TRANSFORMATION_MINI_LAB_RC1
Status	2026 - LEARNER EDITION
Phase	C3-02 Transformation Engineering & Warehousing 14 lessons 34-46 focused hours
Prerequisite	C3-01 Gate PASS. C2B warehouse/modeling foundations may be assumed; implementation depth is taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House rule: analogy -> tiny example -> guided real tool -> break/fix -> independent changed version -> validation
-> production reasoning -> evidence. A successful dbt command alone never proves correct data.



Scenario and architecture

BrightBasket receives customers, orders and order-item feeds. Build a local transformation project that preserves raw evidence, produces current/historical truth, survives a change delivery and hands off tested marts.

Mental model: raw source -> thin staging -> deterministic current/history logic -> dimensions/fact -> serving mart -> tests/docs/handoff. Every arrow has a grain/replay/validation contract.

BUILD ORDER

- 1. Inspect sources and state target grains.
- 2. Build source/staging nodes.
- 3. Build current dimensions + order-line fact; reconcile.
- 4. Add tests; inject one controlled failure and repair.
- 5. Apply order delta and prove same-delta rerun safety.
- 6. Apply customer v2 and prove current + history.
- 7. Generate/inspect docs/lineage where runtime exists.
- 8. Write HANDOFF.md and perform a clean rerun/rebuild where practical.

Competency pass rule

PASS requires M1-M6 all PASS and zero critical failures. No average, percentage or numeric compensation rule exists.

ID	Competency	Required evidence	Status
M1	Layer/source/staging contracts	Raw/source boundaries, target grains and explicit staging contracts are correct.	PASS / REPAIR
M2	Deterministic dimensional/fact build	Dimensions/fact have declared unique grains and reconcile.	PASS / REPAIR
M3	Tests/change safety	Required tests plus one controlled failing test and repair.	PASS / REPAIR
M4	Incremental/replay	BO02 update + BO04 insert apply once; unchanged rerun stable; rebuild path documented.	PASS / REPAIR
M5	History	BC01 bronze->gold retained as legitimate history; current truth correct; no false versions.	PASS / REPAIR
M6	Docs/lineage/handoff	Purpose/grain/dependencies/caveats plus clean-run/recovery handoff.	PASS / REPAIR



Failure/recovery drill

Inject one defect only after a clean baseline is frozen: duplicate dimension key, invalid fact quantity, wrong incremental identity, or broken history change signal. Save failing evidence. Repair the actual cause. Rerun all relevant controls; never disable/weaken a correct test.

EVIDENCE BUNDLE

- Project source/config/SQL/YAML/tests/snapshot/macros used.
- Actual dbt/DuckDB output when runtime available; otherwise exact blocker and static evidence clearly labeled pending.
- Initial/delta/same-delta-rerun identities/counts/measures.
- Current/history evidence.
- Controlled failing test + cause + repair + rerun.
- HANDOFF.md with setup/run/test/rebuild/known limitations.
- 3-5 minute interview defense: grain, incremental identity, history, test, failure, recovery, tradeoff.

GATE LOCK

Completing the Mini-Lab does not pass C3-02. Gate A remains a different domain and stays closed until every Mini-Lab competency passes.